

Kac-Moody Lie Algebras in Deligne Tensor Categories

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Lie Algebras

Throughout this presentation, we assume the field $\mathbb{k}$ we work over is algebraically closed and has characteristic 0.

A *Lie algebra* over a field $\mathbb{k}$ is a vector space L equipped with a *commutator* $[\bullet, \bullet] : L \times L \rightarrow L$ satisfying the following two conditions:

- Alternating:

$$[x, x] = 0.$$

As we assume $\text{char } \mathbb{k} = 0$, this is equivalent to anticommutativity:

$$[x, y] = -[y, x].$$

- Jacobi identity:

$$[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0.$$

For $x \in L$, the *adjoint action* of x is simply the map $\text{ad}_x : y \mapsto [x, y]$.

Examples of Lie Algebras

The *general linear Lie algebra* $\mathfrak{gl}_n$ of all $n \times n$ matrices is a Lie algebra with bracket $[A, B] = AB - BA$. (In fact, any associative algebra becomes a Lie algebra with this bracket.)

As AB and BA always have equal trace, this means that the *special linear Lie algebra* $\mathfrak{sl}_n$ consisting of all $n \times n$ matrices with trace 0 is a Lie algebra with the same bracket.

Motivation for Lie Algebras

Lie algebras are derived from Lie groups. A *Lie group* is a smooth manifold G with a group structure such that the multiplication map

$$m : G \times G \rightarrow G$$

is regular. (This also implies the inverse map $g \mapsto g^{-1}$ is regular.)

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Typically, Lie groups are viewed as a set of *symmetries* of some object. For example, the *general linear group* GL_n is the group of all automorphisms of an n -dimensional vector space. The *special linear group* SL_n is the group of all automorphisms of determinant 1.

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For a Lie group G , the tangent space at the identity $1 \in G$ (i.e., the infinitesimal behavior around 1) is a Lie algebra. For example, GL_n yields $\mathfrak{gl}_n$ this way, and SL_n yields $\mathfrak{sl}_n$, with the correspondence around 1 being given by matrix logarithm/exponentiation.

Modules and the Universal Enveloping Algebra

Given a Lie algebra L , an *L -module* (or *representation*) is a vector space V with an action $\rho : L \rightarrow \text{End}(V)$ such that

$$\rho(x) \circ \rho(y) - \rho(y) \circ \rho(x) = \rho([x, y]).$$

For example, the vector space of $n \times 1$ column vectors is a module over $\mathfrak{gl}_n$ under usual matrix multiplication $A(v) = Av$. Its dual (row vectors) is also a module, but under *negative* multiplication $A(w) = -wA$.

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To formally talk about operators on modules, it is useful to introduce the *universal enveloping algebra* of L . Starting with the tensor algebra $T(L) = \mathbb{k} \oplus L \oplus L^{\otimes 2} \oplus \dots$, we take the quotient

$$\mathcal{U}(L) := T(L) / \langle xy - yx - [x, y] \rangle_{x, y \in L}.$$

This gives us a formal way to multiply elements in L which respects composition of operators on V .

Symmetric Tensor Categories

A *symmetric tensor category* over a field $\mathbb{k}$ is a category $\mathcal{C}$ which is

- $\mathbb{k}$ -linear: all morphism spaces $\text{Hom}(X, Y)$ are $\mathbb{k}$ -vector spaces;
- abelian: has direct sums, kernels, and cokernels;

with a *tensor product* $\otimes : \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$ and a *unit object* $\mathbb{1}$ such that we have

- $\text{End}_{\mathcal{C}}(\mathbb{1}) \cong \mathbb{k}$;
- associativity: $X \otimes (Y \otimes Z) \cong (X \otimes Y) \otimes Z$;
- contraction maps: $\mathbb{1} \otimes X \cong X \cong X \otimes \mathbb{1}$;
- braiding maps: there exist inverse isomorphisms $c_{X,Y} : X \otimes Y \rightarrow Y \otimes X$ and $c_{Y,X} : Y \otimes X \rightarrow X \otimes Y$; and
- rigidity: every object X has a dual X^* .

For example, the category of $\mathbb{k}$ -vector spaces is a symmetric tensor category, with the usual tensor product and $\mathbb{k}$ itself being the unit object.

Algebra and Lie Algebra Objects

In a symmetric tensor category $\mathcal{C}$, we can say certain objects have properties like those of associative algebras or Lie algebras.

An *associative algebra object* is an object $A \in \text{Ob}(\mathcal{C})$ along with an associative *multiplication map* $m : A \otimes A \rightarrow A$ and a *unit map* $u : \mathbb{1} \rightarrow A$.

A *Lie algebra object* is an object $L \in \text{Ob}(\mathcal{C})$ along with a *commutator* $[\bullet, \bullet] : L \otimes L \rightarrow L$ satisfying

- anticommutativity:

$$[\bullet, \bullet] \circ (\text{id} + c_{(12)}) : L \otimes L \rightarrow L$$

is the 0 map.

- Jacobi identity:

$$[\bullet, [\bullet, \bullet]] \circ (\text{id} + c_{(123)} + c_{(132)}) : L \otimes L \otimes L \rightarrow L$$

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Representation Theory of $\mathfrak{gl}_n$

Recall that the n -dimensional vector space V and its dual V^* are both representations of $\mathfrak{gl}_n$. Also, $\mathfrak{gl}_n \cong V \otimes V^*$.

The following four maps are our “building blocks”:

- $\text{id}_V : V \rightarrow V$
- $\text{id}_{V^*} : V^* \rightarrow V^*$
- $\text{ev} : V^* \otimes V \rightarrow \mathbb{k}$ given by dot product (or trace)
- $\text{coev} : \mathbb{k} \rightarrow V \otimes V^*$ mapping 1 to the identity matrix

By taking tensors, direct sums, kernels/cokernels, etc., we can recover almost all of the representation theory of $\mathfrak{gl}_n$.

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n itself plays a role in the composition rule: $\text{ev} \circ \text{coev} : \mathbb{k} \rightarrow \mathbb{k}$ is $n \cdot \text{id}_{\mathbb{k}}$.

Deligne Categories

Following this construction but replacing n with a complex parameter δ gives us the *Deligne category* $\underline{\text{Rep}}(\text{GL}_\delta)$, a natural interpolation of the representation theory of $\mathfrak{gl}_n$ to complex parameters. V mimics a vector space of “dimension” δ .

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The object $\mathfrak{gl}_\delta := V \otimes V^*$ still has a multiplication map:

$$(V \otimes V^*) \otimes (V \otimes V^*) = V \otimes (V^* \otimes V) \otimes V^* \xrightarrow{1 \otimes \text{ev} \otimes 1} V \otimes \mathbb{1} \otimes V^* \cong V \otimes V^*.$$

Thus, $\mathfrak{gl}_\delta$ is a Lie algebra object interpolating $\mathfrak{gl}_n$.

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We also still have the action of $\mathfrak{gl}_\delta$ on V :

$$(V \otimes V^*) \otimes V = V \otimes (V^* \otimes V) \xrightarrow{1 \otimes \text{ev}} V \otimes \mathbb{1} \cong V,$$

and likewise on V^* .

Generalized Cartan Matrix

The construction we aim to generalize is a Kac-Moody Lie algebra.

We start with a *generalized Cartan matrix* A , which is an $n \times n$ matrix with integer entries $a_{i,j}$ such that

- $a_{i,i} = 2$,
- $a_{i,j} \leq 0$ for $i \neq j$, and
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For our purposes, the matrix A is always derived from a (connected simply-laced) diagram. Starting with an undirected connected graph with vertices labeled 1 through n , we define entries

- $a_{i,i} = 2$,
- $a_{i,j} = -1$ for i and j neighbors, and
- $a_{i,j} = 0$.

In particular, such a matrix is always symmetric.

Realization

A *realization* of $A = (a_{i,j})$ is a triple $(\mathfrak{h}, \Pi, \Pi^\vee)$ where

- $\mathfrak{h}$ is a complex vector space of dimension $2n - \text{rank}(A)$,
- $\Pi = \{\alpha_1, \dots, \alpha_n\} \subseteq \mathfrak{h}^*$ is a linearly independent set of *simple roots*, and
- $\Pi^\vee = \{\alpha_1^\vee, \dots, \alpha_n^\vee\} \subseteq \mathfrak{h}$ is a linearly independent set of *simple coroots* such that $\alpha_j(\alpha_i^\vee) = a_{i,j}$.

Kac-Moody Lie Algebra

Given a realization $(\mathfrak{h}, \Pi, \Pi^\vee)$, the *Kac-Moody Lie algebra* $\mathfrak{g}(A)$ is generated by

- the *Cartan subalgebra* $\mathfrak{h}$;
- the *positive Chevalley generators* $e_1, \dots, e_n$; and
- the *negative Chevalley generators* $f_1, \dots, f_n$

defined by the following relations: for $h, h' \in \mathfrak{h}$,

- $[h, h'] = 0$,
- $[h, e_i] = \alpha_i(h)e_i$,
- $[h, f_i] = -\alpha_i(h)f_i$,
- $[e_i, f_i] = \alpha_i^\vee \in \mathfrak{h}$,
- $[e_i, f_j] = 0$ for $i \neq j$,
- $(\text{ad}_{e_i})^{1-a_{i,j}}(e_j) = 0$ for $i \neq j$, and
- $(\text{ad}_{f_i})^{1-a_{i,j}}(f_j) = 0$ for $i \neq j$.

The last two relations are called *Serre relations*.

Properties of Kac-Moody Lie Algebras

$\mathfrak{g}(A)$ generalizes properties of finite-dimensional Lie algebras.

- $\mathfrak{g}(A)$ decomposes as $\mathfrak{g}(A) = \mathfrak{n}_- \oplus \mathfrak{h} \oplus \mathfrak{n}_+$ where $\mathfrak{n}_+$ is generated by e_i 's and $\mathfrak{n}_-$ by f_i 's, both defined solely by Serre relations.

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- $\mathfrak{g}(A)$ has a *Cartan involution* defined by

$$\omega(e_i) = -f_i, \quad \omega(f_i) = -e_i, \quad \omega(h) = -h \quad \text{for } h \in \mathfrak{h}.$$

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- $\mathfrak{g}(A)$ has a bilinear form $(\bullet, \bullet) : \mathfrak{g}(A) \times \mathfrak{g}(A) \rightarrow \mathbb{C}$ which is *invariant*:

$$(x, [y, z]) = ([x, y], z).$$

In the finite-dimensional case (e.g., $\mathfrak{gl}_n$), this is the trace form.

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- There is a *Casimir operator* Ω which is a quadratic element of $\mathcal{U}(\mathfrak{g}(A))$ which commutes with $\mathfrak{g}$. In the finite-dimensional case,

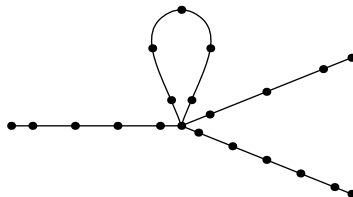
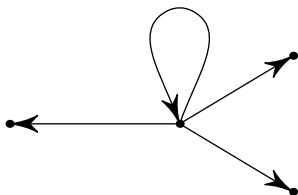
$$\Omega = \sum_i u_i u^i$$

for dual bases $\{u_i\}$ and $\{u^i\}$ under the invariant bilinear form.

Associated Diagram

Our goal is to construct the Kac-Moody Lie algebra as an ind-object in a tensor product of Deligne categories and recover its properties.

Start with a weakly connected directed graph D , possibly with multiple edges and self-loops. For each edge $i \in \mathbf{E}(D)$, we pick an integer parameter $n_i \geq 2$. From D and n_i 's, we construct a new graph by inserting $n_i + 1$ consecutive vertices along i , which we call the *associated diagram*.



Kac-Moody Lie Algebras along One Edge

We can consider the Kac-Moody Lie algebra $\mathfrak{g}$ corresponding to the associated diagram. The diagram for $\mathfrak{gl}_n$ consists of $n - 1$ vertices in a chain, so the subalgebra of $\mathfrak{g}$ corresponding to a single edge i looks like a copy of $\mathfrak{gl}_{n_i+4}$:

$$\begin{array}{ccccccc}
 v & & v_{i,s} & & (n_i - 1) \text{ additional vertices} & & v_{i,t} & & w \\
 \bullet & \text{---} & \bullet & \text{---} & \bullet & \text{---} & \bullet & \text{---} & \bullet
 \end{array}$$

$$\left(\begin{array}{cc|cc|cc}
 1_v & e_v & & & & \\
 f_v & 1_{i,s} & e_{i,s} & & & \\
 & f_{i,s} & & & & \\
 & & & \mathfrak{gl}_{n_i} & & \\
 & & & & e_{i,t} & \\
 & & & & f_{i,t} & 1_{i,t} & e_w \\
 & & & & f_w & 1_w &
 \end{array} \right)$$

Note that $e_{i,t}$ generates a copy of the n_i -dimensional vector space V_i , and $e_{i,s}$ generates a copy of V_i^* . Both e_v and e_w just generate $\mathbb{1}$.

Categorical Kac-Moody Lie Algebras

Replace each parameter n_i with a complex parameter δ_i . We work in the tensor product of Deligne categories $\underline{\text{Rep}}(\text{GL}_{\delta_i})$ for $i \in \mathbf{E}(D)$, which contains objects $\mathfrak{gl}_{\delta_i}$, V_i , and V_i^* .

We construct a Lie algebra ind-object $\text{KM}(D, \delta) = \mathfrak{u}_- \oplus \mathfrak{l} \oplus \mathfrak{u}_+$ where

$$\mathfrak{l} = \left(\bigoplus_{i \in \mathbf{E}(D)} V_i \otimes V_i^* \right) \oplus \left(\bigoplus_{i \in \mathbf{E}(D)} (\mathbb{1}_{i,s} \oplus \mathbb{1}_{i,t}) \right) \oplus \left(\bigoplus_{v \in \mathbf{V}(D)} (\mathbb{1}_v \oplus \mathbb{1}'_v) \right),$$

$$\mathfrak{u}_+ = FL \left(\bigoplus_{v \in \mathbf{V}(D)} \mathbb{1}_{v,+} \oplus \bigoplus_{i \in \mathbf{E}(D)} (V_{i,s}^* \oplus V_{i,t}) \right) / \langle \text{Serre relations} \rangle,$$

and $\mathfrak{u}_-$ is defined analogously.

Along a single edge i , $\text{KM}(D, \delta)$ exhibits the same behavior as in the previous slide.

Properties of the Categorical Kac-Moody Lie Algebra

- We have a monoidal functor F from the tensor product of Deligne categories to itself that interchanges the tautological object V_i of dimension t_i with its dual. Then $KM^* = F(KM)$ will have an induced bracket and be a Lie algebra object.

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- We constructed a Lie algebra homomorphism $\omega : \mathrm{KM} \rightarrow \mathrm{KM}^*$ that satisfies $F(\omega) = \omega^{-1}$ to generalize the Cartan involution.
- We defined a pairing $(\bullet, \bullet) : \mathrm{KM} \otimes \mathrm{KM} \rightarrow \mathbb{1}$ on the generators of the categorical Lie algebra and extended it through invariance. This generalizes the invariant bilinear form.
- We computed the Casimir operator Ω as a sum of degree ≤ 2 elements of $\mathcal{U}(\mathrm{KM})$ that commutes with the generators of KM , generalizing the same notion for Kac-Moody Lie algebras.

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References

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